Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Atwater, James E		
Project Title:	Advanced Magnetic Methods for the Separation and Gasification of Solid Wastes in Microgravity and Hypogravity		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Myrtle Creek	**State:**	OR
Zip Code:	97457-0102	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	03/01/2003	**End Date:**	01/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Akse, James R. (UMPQUA Research Company) Jovanovic, Goran N. (Chemical Engineering Dept., Oregon State University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this project, the development of microgravity compatible methods for solid waste decomposition continues. For the next three years the work will focus on advanced magnetic materials development, development of improved magnetically assisted filtration and fluidization techniques, reaction engineering for the decomposition of solid wastes, extensive theoretical development of the associated processes, and the development of mathematical models to facilitate numerical simulations. Develop microgravity and hypogravity compatible methods for separation and conversion of solid wastes into useful products. Improved methods of solid waste treatment are needed to support NASA's Solid Waste Mangement goals related to long duration manned missions. Hypogravity and Microgravity compatibility issues are especially critical. The conversion of wastes to resources is very beneficial.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will Advance the fundamental state of knowledge of Magnetically Stabilized Fluidized Bed operations in both microgravity and normal gravity conditions. This will have a direct impact in the expansion of capabilities which may be employed by the Chemical Industry, Biotechnology, and Materials Science Sectors. State of the art CFD-DPM fluidization modeling software will provide a valuable tool for the industrial and research fluidization communities.
Task Progress:	The development of improved solid waste management methods and equipment is a critical requirement for long duration manned missions such as a Mars transit, Mars base, or lunar outpost. Methods for solid waste stabilization and treatment must be capable of operating in microgravity and hypogravity environments. Investigators at UMPQUA Research Company (URC) and Oregon State University (OSU) are working toward the development of microgravity and hypogravity capable methods for the gasification of solid wastes, which employ magnetic forces as a substitute for gravity. Properly designed magnetic field gradients are used to control the degree of fluidization, or conversely consolidation, of granular ferromagnetic media in a variety of gravitational environments. These methods are directly applicable to the decomposition of organics within a variety of solid waste materials via fluidized bed reactor operations. The separation and concentration of solids using magnetically regenerable filter elements has also been demonstrated. Both of these techniques may be applied to either microgravity or hypogravity operations. The solid waste separation and gasification methods under development offer direct benefit to both Advanced Life Support and In Situ Resource Utilization communities. Gasification of solid wastes results in the conversion of highly heterogeneous material into inorganic residue and a variety of gases. This minimizes the infrastructure required for long-term waste storage and/or disposal. Similarly, the requirement for stabilization with respect to the growth of pathogens is minimized. Gasification reactions may be employed to recover a wide range of potentially valuable resources, including: carbon dioxide, water, methane, hydrogen, and oxygen, using selective reaction scenarios optimized to meet mission specific resource recovery requirements. In previous work, we have focused on the development of novel electromagnets and related magnet design software, development of novel high temperature compatible ferromagnetic catalysts, development of magnetic control methods to achieve magnetically assisted fluidization and filtration unit operations, microgravity flight experiments providing the first demonstration of stable fluidization in the weightless environment, and development of mathematical models describing these processes under conditions of variable magnetic field strength and variable gravity. During the previous twelve month period, we have concentrated our efforts on the investigation of a variety of gasification reaction scenarios and on the development of a three-dimensional computational fluid dynamic – discrete particle model (CFD-DPM) describing magnetically assisted gasification (MAG) related processes. These are described in greater detail in the 2nd Annual Progress Report. Our study of gasification reactions covered a wide range of reaction conditions, including: steam reforming, catalytic steam reforming, steam reforming with oxygen injection (with and without catalysis), pyrolysis, catalytic pyrolysis, and incineration using both air and oxygen. This work has identified suitable conditions for the complete conversion of solids to inorganic ash and gaseous byproducts. Different reaction scenarios have also been identified to maximize the production of hydrogen, oxygen, carbon dioxide and water. The direct benefit of oxygen reclamation is clear. Hydrogen recovery from solid wastes can be employed to minimize the requirement for supply (and/or resupply) for use with Sabatier CO2 reduction reactors and related systems for the recovery of water from lunar regolith.
Bibliography Type:	Description: (Last Updated: 04/05/2010)
Abstracts for Journals and Proceedings	Atwater JE, Akse JR, Wheeler RR Jr, Jovanovic GN, Pinto-Espinoza R, Sornchamni T, Reed BP. "Magnetic Control of Granular Ferromagnetic Media in Microgravity for Advanced Life Support Applications." Habitation 2004 Conference, Orlando, FL., January 4-7, 2004. Habitation 2004 Conference abstracts. Orlando, FL., January 4-7, 2004. , Jan-2004
Abstracts for Journals and Proceedings	Jovanovic GN, Reed BP, Cruz-Fierro C, Atwater JE. "Influence of Interparticle Forces on Liquid Solid Fluidized Beds: Experiments and Simulations (CFD-DPM)." AIChE Annual Meeting, Austin, TX., November 7-12, 2004. AIChE Annual Meeting. 2004 Nov 7-12; Paper 282b. , Nov-2004
Abstracts for Journals and Proceedings	Cruz-Fierro CF, Reed BP, Jovanovic GN. "Interpolation of Voidage in CFD-DPM Simulation of Liquid-Solid Fluidized Beds." AIChE Annual Meeting, Austin, TX., November 7-12, 2004. AIChE Annual Meeting. 2004 Nov 7-12; Paper 282e. , Nov-2004
Articles in Peer-reviewed Journals	Jovanovic GN, Sornchamni T, Atwater JE, Akse JR, Wheeler RR Jr. "Magnetically assisted liquid–solid fluidization in normal and microgravity conditions: experiment and theory." Powder Technology. 2004 Nov 11;248(2-3):80-91. http://dx.doi.org/10.1016/j.powtec.2004.09.028 , Nov-2004
Articles in Peer-reviewed Journals	Sornchamni T, Jovanovic GN, Reed BP, Atwater JE, Akse JR, Wheeler RR Jr. "Operation of magnetically assisted fluidized beds in microgravity and variable gravity: experiment and theory." Advances in Space Research. 2004 Oct;34(7):1494-8. http://dx.doi.org/10.1016/j.asr.2004.07.010 , Oct-2004